PAPER NAME: BASIC UNIVERSAL HUMAN ENGINEERING
PAPER CODE: COE4312
APPLICABLE FOR DEPARTMENT: AEIE, CSE, ECE, EE, BT, ME, CE, CHE, IT

GROUP – A

MCQ *(Choose the Correct Alternative)*

Module 1

(i) What are the basic desires of every human being for which they are working?
(a) Physical facilities
(b) Realization and understanding Identifier
(c) Happiness and prosperity
(d) Continuous happiness and prosperity

(ii) Needs of the body are------.
(a) Continuous
(b) Temporary
(c) Both (A) and (B)
(d) None of the above

(iii) When we participate in the larger order by aligning and exercising our aspirations, this participation can be appreciated by us through _____.
(a) Right Understanding
(b) Happiness and Prosperity
(c) Ethical values
(d) Desire for physical facilities

(iv) Value Education is important for students because it helps students to correctly identify
(a) Key to success
(b) Aspirations
(c) Acceptance
(d) Values

(v) Which of the following sequence is divergent?
(a) $\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5} \ldots to\ \infty\right\}$.
(b) $\left\{\left(1+\frac{1}{n}\right)^n\right\}, n \in N$
(c) $\left\{\frac{\sin n\pi}{n}\right\}, n \in N$
(d) $\{2^n\}, n \in N$.

(vi) The sequence $\left\{\frac{1}{3^n}\right\}, n \in N$ is
(a) monotonic increasing
(b) divergent
(c) oscillatory
(d) monotonic decreasing.

(vii) The value of the constant a, so that the vector $\vec{f} = (x + 2y + az)\hat{\imath} + (2x - 3y - z)\hat{\jmath} + (4x - y + 2z)\,\hat{k}$ is irrotational, is

(a) 1
(b) 2
(c) 0
(d) 4.

(viii) If $\varphi(x, y, z) = x^2y + y + z^2$ then grad φ at $(1, -1, 1)$ is
(a) $2(\hat{\imath} - \hat{\jmath} + \hat{k})$
(b) $2(\hat{\imath} + \hat{\jmath} - \hat{k})$
(c) $2(-\hat{\imath} + \hat{\jmath} + \hat{k})$
(d) $3(\hat{\imath} - \hat{\jmath} + \hat{k})$.

Module 2

(i) The resonant frequency of an electrical LCR circuit is given by
(a) $\nu = 2\pi\sqrt{LC}$
(b) $\nu = \frac{1}{2\pi\sqrt{LC}}$
(c) $\nu = \frac{2\pi}{\sqrt{LC}}$
d) $\nu = \frac{1}{2\pi}\sqrt{\frac{L}{C}}$

(ii) A damped oscillator is acted upon by an external force $F = F_0 cos\omega t$. If ω_0 is the natural frequency of oscillation, the frequency of oscillation of the system at steady state is
(a) ω_0
(b) ω
(c) $\omega - \omega_0$
(d) $\omega + \omega_0$

(iii) Damping constant γ resembles the dimension of
(a) length
(b) mass
(c) time
(d) frequency

(iv) A function $x(t) = Asin^2\omega_o t$ represents
(a) SHM about a point $x = \frac{A}{2}$ with time period $\frac{\pi}{\omega_0}$
(b) SHM about a point $x = \frac{A}{2}$ with time period $\frac{2\pi}{\omega_0}$
(c) SHM about a point $x = 0$ with time period $\frac{\pi}{\omega_0}$
(d) SHM about a point $x = 0$ with time period $\frac{2\pi}{\omega_0}$

(v) The value of drain current when V_{gs}=pinch off voltage is
(a) 1mA
(b) 0
(c) 2mA
(d) 0.5A

(vi) For an n-channel FET, the direction of drain current flow is
(a) Gate to source
(b) Source to drain

(c) drain to source
(d) Drain to Gate

(vii) Comparing the size of BJT and FET, choose the correct statement
(a) Both have same size
(b) FET is larger than the BJT
(c) BJT is larger than the FET
(d) Depends on same application

(viii) The drain current in JFET is due to
(a)flow of Minority carriers
(b) flow of Majority carriers
(c) flow of both minority and majority carriers
(d) none of the above

Module 3

(i) Suppose both p1 and p2 are pointers to the same array. Now which of the following operations on the pointers are not allowed in C language?
(a) p1 + p2
(b) p1 - p2
(c) p1+1
(d) p2++

(ii) What is the default return type of malloc( )?
(a) int
(b) (int*)
(c) (void*)
(d) void

(iii) Which of the following statements is FALSE?
(a) Array bound is not checked in C
(b) Local variable gets precedence over global variable
(c) Garbage collection is not automatic in C
(d) main() is a function with no default return type.

(iv) What will the output be?

```
int n = 10;
void fun (int x)
{
        static int r = x;
        if(r < 2)
        {
                printf("%d %d ", n, r);
                fun(++ r);
        }
}

int main (void)
{
        static int n;
        fun(n);
```

```
        printf("%d",n);
}
```

(a) 10
(b) 10 0 10 10
(c) 100 10 1 1
(d) 10 0 10 1 10

(v) In Thevenin's theorem, to find the Thevenin's equivalent resistance:
(a) All independent current sources are short circuited and independent voltage sources are open circuited.
(b) All independent voltage and current sources are open circuited.
(c) All independent voltage and current sources are short circuited.
(d) All independent voltage sources are short circuited and all independent current sources are open circuited.

(vi) The superposition theorem is applicable to:
(a) current only.
(b) voltage and power.
(c) voltage and current.
(d) voltage, current and power.

(vii) An ideal current source should possess
(a) zero source resistance
(b) infinite source resistance
(c) large value of current
(d) small value of current

(viii) The number of equations required to solve network by mesh analysis is equal to number of
(a) nodes
(b) independent loops
(c) branches
(d) source

Module 4

(i) SI unit of Strain is
(a) millimetre
(b) metre
(c) inch
(d) dimensionless

(ii) According to the Hook's law
(a) stress is directly proportional to strain
(b) stress is inversely proportional to strain
(c) stress is square root of strain
(d) none of the above

(iii) The ratio of lateral strain and longitudinal strain
(a) stress
(b) strain
(c) Poisson's ratio
(d) modulus of elasticity

(iv) Shear stress is quantified by
(a) unit angular deformation
(b) unit longitudinal deformation
(c) unit lateral deformation
(d) none of the above

(v) What is the return type of the fopen() function in C?
(a) char *
(b) bool
(c) FILE *
(d) void *

(vi) What is the function feof() used for in C?
(a) To close a file
(b) To write to a file
(c) To check if the end of file has been reached
(d) To flush the file buffer

(vii) How is the size of a structure determined in C?
(a) By the total size of its members
(b) By the total size of its largest member
(c) By the total size of its smallest member
(d) By the count of its members

(viii) Which one of the following is correct syntax for opening a file?
(a) FILE *fopen(const *filename, const char *mode)
(b) FILE *fopen(const *filename)
(c) FILE *open(const *filename, const char *mode)
(d) FILE open(const*filename)

FILL IN THE GAPS

1. The IEEE Single Precision representation for the decimal number $(65.5)_{10}$ is _____.

2. The output of the following program is _____.

```
int main(void)
{
    const int a = 10;
    int *p = &a;
    *p = 20;
    printf("%d",a);
    return 0;
}
```

3. The output of the following program is _____.

```
int x = 2, y = 6, z = 1;
while(z > 0)
{
    z= (++y + z) % x ;
    printf("%d ",z);
```

}

4. $(24562.76)_8$ = (_____)$_{16}$
5. The Decimal value for 1 10000001 01000…00 represented in IEEE 754 32-bit pattern is _____.
6. At resonance the current through a RLC series circuit will be _____.
7. For a series RLC circuit the current at resonance is _____ .
8. The peak factor of pure sinusoidal waveform is _____ .
9. A shunt generator is running at 1000 rpm and has a generated emf of 200V. If the speed increases to 1200 rpm the generated emf will be nearly_____volt.
10. In a three-phase system, the line emf's are_____degree apart.
11. For reversing the direction of d.c. motor_____.
12. The plane of maximum obliquity is inclined to the major principal plane at an angle _____$+\frac{\phi_{max}}{4}$
13. The basic perfect frame is _____ (parallelogram/rectangle/triangle).
14. A hollow shaft will transmit _____ (more/less) power than a solid shaft of same weight and material.
15. The Rankine's constant for mild steel is _____.
16. For a circular column having it's ends hinged, the slenderness ratio is 160. The *l/d* ratio of the column is ____.

GROUP – B

2. (a) Explain various points of a stress-strain curve of mild steel.

[(CO1)(Understand/LOCQ)]

(b) Draw the stress – strain relationship on a graph paper using the table given below. Identify whether the material is ductile or brittle. Identify proportional limit, yield stress, plastic zone, ultimate stress and breaking stress from the stress – strain curve.

Strain	Stress (N/mm^2)
0	0
0.001	10
0.002	20
0.003	30
0.004	40
0.005	38
0.006	37
0.007	64
0.008	70
0.009	65
0.01	52

[(CO2)(Apply/IOCQ)]

3 + 9 = 12

3. (a) Find the centroid of the lamina shown in Fig. 2 (All dimensions are in mm.)

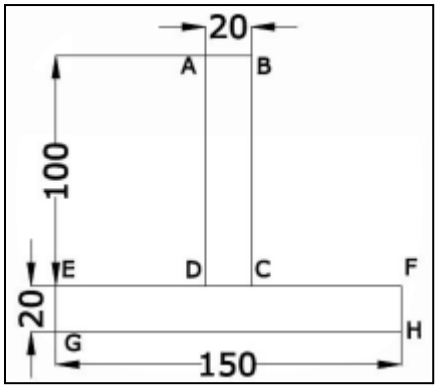

Fig. 2

[(CO3)(Analyse/HOCQ)]

(b) Determine the value of moment of inertia of the section mentioned in Fig. 2 along both the axes. *[(CO4)(Remember/LOCQ)]*

6 + 6 = 12

4. (a) A member ABCD is subjected to point loads P_1,P_2, P_3 and P_4 as shown in the Fig. 7 below.

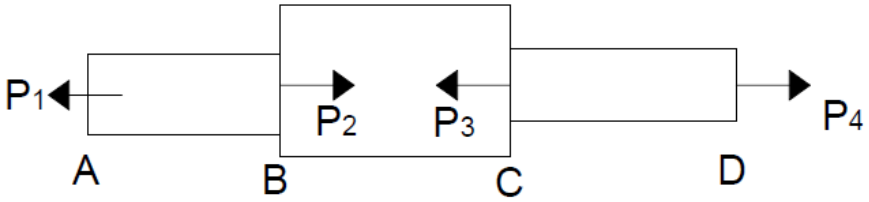

Fig. 7

The area and lengths of respective parts are given below:

(a) AB (area= 625 mm^2 and length =120cm.)
(b) BC (area= 2500 mm^2 and length =60 cm.)
(c) CD (area= 1250 mm^2 and length =90 cm.)

Calculate the forces necessary for equilibrium if P_1= 45kN, P_3= 450kN and P_4=130 kN. Determine the total elongation of the member, assuming E= 2.1 X 10^5 N/mm^2 *[(CO1)(Analyse/HOCQ)]*

(b) A tensile load of 50 kN is acting on a rod as shown in Fig. 8 of 50 mm diameter and length 5m. Determine the length of a bore of 25 mm that can be made central in the rod, if the total extension is not to exceed by 25% under same tensile load. Take E=2.05 x10^5 N/mm^2.

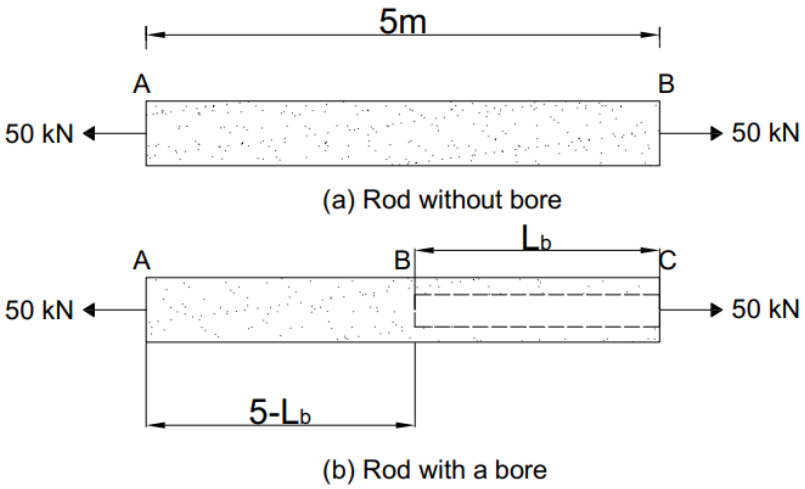

Fig. 8

[(CO1)(Analyze/HOCQ)]

6 + 6 = 12

5. (a) How compiler and interpreter are different? *[(CO1)(Understand/LOCQ)]*
   (b) State the difference between volatile and non-volatile memory with proper example. *[(CO1)(Remember/LOCQ)]*
   (c) Create a flowchart to determine the minimum number of currency notes needed to make up a specified amount using denominations of 100, 50, 20, 10, 5, 2, and 1 rupee notes. *[(CO3)(Apply/IOCQ)]*

3 + 4 + 5 = 12

6. (a) Imagine a signed integer variable is represented by 3 bits. Then what is the range of numbers accepted by this variable. *[(CO2)(Analyse/HOCQ)]*
   (b) Briefly describe the functions of different components of a conventional digital computer with a suitable block diagram. *[(CO1)(Remember/LOCQ)]*
   (c) Draw a flowchart to print the largest of three numbers.

[(CO3)(Apply/IOCQ)]

3 + 5 + 4 = 12

7. (a) Convert the following numbers from one number system to the specified one:
   (i) $(0.865)_{10} = (?)_8$
   (ii) $(4D.21)_{16} = (?)_{10}$
   (iii) $(24.36)_8 = (?)_{10}$
   (iv) $(1111.1101)_2 = (?)_{10}$ *[(CO2)(Apply/IOCQ)]*
   (b) Consider the following declaration of a variable in your C program: float x=12.75; Show the 32 bit pattern that will represent x in the memory, using
   IEEE754 floating point representation. *[(CO2)(Apply/IOCQ)]*

(2 + 2 + 2+2) + 4 = 12

8. (a) Draw a flowchart to check whether a given number is prime or not.

[(CO3)(Apply/IOCQ)]

   (b) Using 8 bit number representation, add $+77_{10}$ to -77_{10} using 1's complement. Perform the same task using 2's complement representation.

[(CO2)(Apply/IOCQ)]

   (c) Perform the following operations while showing all intermediate steps.
   i) Convert $(FF58.45)_{16}$ to decimal number system
   ii) Convert $(143.52)_8$ to hexadecimal number system

[(CO2)(Apply/IOCQ)]

4 + (2 + 2) + (2 + 2) = 12

9. (a) Consider the following declaration of a variable in your C program:
   float x=4.5;
   Show the 32 bit pattern that will represent x in the memory, using IEEE754 floating point representation. *[(CO2)(Apply/IOCQ)]*

(b) Find r such that $(121)_r = (144)_8$, where r and 8 are the base/radix of the number system. *[(CO1,CO2)(Apply/IOCQ)]*

(c) Draw a flowchart to find the Greatest Common Divisor (GCD) of two integers. *[(CO3)(Apply/IOCQ)]*

4 + 4 + 4 = 12

10. (a) What is meant by peritactic reactions for austenite? What is hypoeutectoid steel? *[(CO3)(Remember/LOCQ)]*

(b) In Fe-C phase diagram, liquid solution with austenite is with the carbon percentage of 2.14% (wt), while with the α-ferrite it is only 0.022% (wt). Elaborate the reason with proper reasons. *[(CO3)(Analyse/HOCQ)]*

(c) What is meant by lower and upper critical temperature in Fe-C phase diagram? *[(CO3)(Remember/LOCQ)]*

(2 + 2) + 5 + 3 = 12

11. (a) Find out the mole fraction of Schottky defect in NaCl crystal at 2900 K, when ΔH for Schottky is 2 eV, 1 eV=1.608 x 10^{-19} J and k_B = 1.38 x 10^{-23} J/K *[(CO1)(Apply/IOCQ)]*

(b) Why the conduction band and valence band appear in a compound? Elaborate with proper diagram. *[(CO2)(Remember/LOCQ)]*

(c) A semiconductor diode laser has a peak emission wavelength of 1.55 μm. Find its band gap in eV. The velocity of light is 3 x 10^8 m/s, h=6.63 x 10^{-34} J-s. *[(CO2)(Apply/IOCQ)]*

5 + 4 + 3 = 12

12. (a) State and prove maximum power transfer theorem. Also state the value of efficiency at which maximum power is transferred from source to load. *[(CO1) (Remember/LOCQ)]*

(b) An iron ring of 10cm^2 cross section and 100 cm mean circumference has a 0.2 cm saw-cut made in it. Calculate the current required in a magnetising coil having 1000 turns to produce flux of 1mWb in the air gap. Iron is such that when flux density is 1 Wb/m^2, the relative permeability μ_r=400.Consider no leakage is present. μ_0= 4π × 10^{-7} H/m, *[(CO3) (Evaluate/HOCQ)]*

(2 + 4 + 1) + 5 = 12

13. (a) What do you mean by composition and resolution of a vector? *[(CO1)(Remember/LOCQ)]*

(b) State and explain Parallelogram Law of forces? *[(CO2)(Remember/LOCQ)]*

(c) A force of 100 N is acting at a point A as shown in Fig. 2. Determine the moments of this force about O.

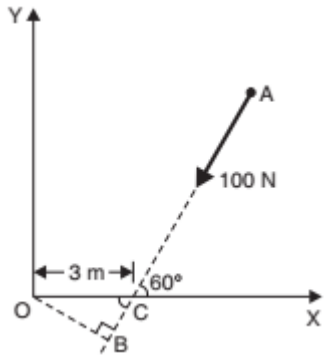

Fig. 2

[(CO2)(Apply/IOCQ)]

4 + 4 + 4 = 12

GROUP – C

14. (a) Find out the maximum and minimum of two numbers using ternary operator. *[(CO3)(Apply/LOCQ)]*

(b) Briefly explain about the basic data types that C language supports. *[(CO2, CO3)(Remember/LOCQ)]*

(c) Write a C program to print all the prime numbers between 1 to n, where n is a user input. *[(CO3, CO5)(Apply/IOCQ)]*

3 + 4 + 5 = 12

15. (a) What is the difference between a float and a double data type in C? *[(CO3)(Analyse/IOCQ)]*

(b) Discuss the concept of switch-case fall-through. How can you avoid it? *[(CO3)(Remember/LOCQ)]*

(c) Write a C program that simulates the behaviour of a basic ATM system using switch statements, allowing the user to withdraw, deposit, and check balance.

[(CO2, CO3, CO5)(Apply/IOCQ)]

4 + (2+2) + 4 = 12

16. (a) A resistance of 16 Ω , inductance of 0.15 H and a capacitance of 130μF are connected in series across a 100 V, 50 Hz supply. Determine:

(i) the impedance of the circuit
(ii) the current
(iii) power factor
(iv) active, reactive and apparent power of the circuit
(v) voltage across each element
(vi) resonant frequency or the frequency at which maximum current flows
(vii) circuit current at resonance or the amount of maximum current
(viii) Draw the phasor diagram

[(CO4) (Evaluate/HOCQ)]

(b) A parallel circuit consists of two branches, one consisting of a noninductive resistor of 150Ω in series with an inductor of inductance 318μH, the other

consisting of a noninductive resistance of 300Ω in series with a capacitor of 0.01μF capacitance. The circuit is connected across a 25V, 100 kHz supply. Determine (i) the current in each branch, (ii) the total current, (iii) overall power factor, (iv) the total impedance of the circuit.

[(CO4) (Analyse/HOCQ)]

8 + 4 = 12

17. (a) Derive the relationship between the line voltage and phase voltage in a star connected and delta conected 3 phase circuit

[(CO4) (Remember/LOCQ)]

(b) Derive the torque equation of a DC motor?

[(CO2) (Remember/LOCQ)]

(c) An 8 pole lap connected armature has 40 slots with 12 conductors per slot, generating a voltage of 5000 Volts. Determine the speed at which it is running if the flux per pole is 50mWb. Also calculate the terminal voltage at full load of 20A if the armature resistance is 0.5 Ω and the machine is running as a shunt generator having a shunt field resistance of 250 Ω.

[(CO2) (Analyse/HOCQ)]

3 + 4 + 5 = 12

18. (a) A 50 kVA, 2400/120V, 50 Hz single phase transformer has the following test results:

O.C. Test (Low voltage side):	120 V	9.65 A	396 W
S.C. Test (High voltage side):	92 V	20.8 A	810 W

Determine the parameters of the equivalent circuit of the transformer and the efficiency of the transformer at half load and 0.85 power factor lagging.

[(CO5) (Apply/IOCQ)]

(b) A 3-phase, 4 pole, 50 Hz induction motor has a slip of 1% at no load and 3% at full load. Determine (i) synchronous speed, (ii) no-load speed, (iii) full load speed, (iv) frequency of rotor current at full load, (v) frequency of rotor current at standstill.

[(CO6) (Evaluate/HOCQ)]

7 + 5 = 12

19. Evaluate the value of deflection and rotation at the free end of the cantilever beam shown in Fig. 30 using conjugate beam method. Assume EI to be constant throughout.

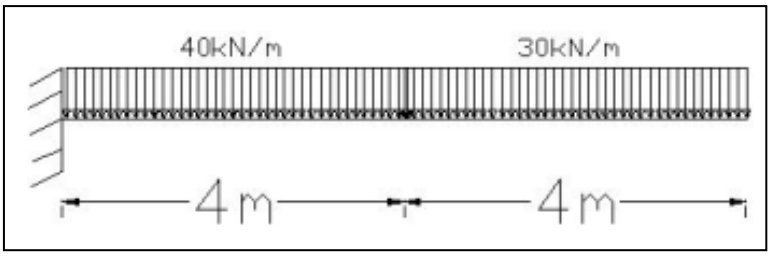

Fig. 30

[(CO3)(Analyse/HOCQ)]

12

20. A beam of span 30m carries a point load as shown in Fig. 31. The moment of inertia (I) of the beam is assumed to be constant throughout. Evaluate slope and deflection at the free end and at the midpoint. Use conjugate beam method. Assume material to be uniform throughout.

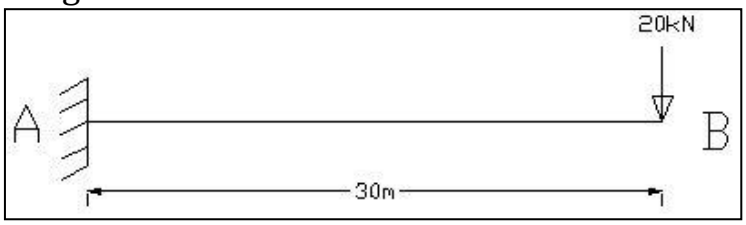

Fig. 31

[(CO3)(Analyse/HOCQ)]
12

21. (a) Define Bulk Modulus. Deduce the relation between Young's Modulus, Bulk Modulus and Poisson's ratio.

[(CO5)(Remember/LOCQ)]

(b) An aluminium specimen shown in Fig. 4 has a diameter of d_0 = 25 mm and a gauge length of L_0 = 250 mm. If a force of 150 kN elongates the gauge length 1.20 mm, determine the modulus of elasticity. Also, determine by how much the force causes the diameter of the specimen to contract. Take G_{al} = 26 GPa and σ_y= 440 MPa.

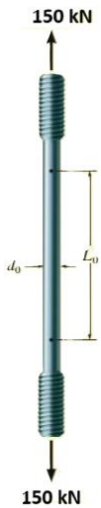

Fig. 4

[(CO5)(Apply/IOCQ)]
(3 + 3) + (3 + 3) = 12

22. (a) A program has been written for generation of random DNA sequences. It is expected that there is a 25% sequence identity between pairs of random sequences in gapped alignments. However when the alignment related calculation is done, more than 25% sequence identity is obtained. Explain the difference in the identity value.

[(CO2)(Analyze/IOCQ)]

(b) Out of the two protein sequences of catalase one sequence is of *Vibrio cholerae* and another of *Drosophilla melanogastor.* Mention the approaches to be followed that reveal similarity among themselves.

[(CO2) (Understand/LOCQ)]

(c) Mention the role of gap in sequence alignment.

[(CO2) (Understand/LOCQ)]

(d) The default values of opening gap penalty is much higher and extention gap penalty – state why this type of discrimination is found.

[(CO2)(Analyze/IOCQ)]

(e) In another situation if you are given one sequence what technique will you be using to find out its relationship with others?

[(CO2)(Consider/HOCQ)]

3 + 2 + 2 + 2 + 3 = 12

23. (a) Explain the use of ON DELETE CASCADE in SQL.

[(CO4)(Understand/LOCQ)]

(b) Write the following queries using basic relational algebra operations:
Consider the following database
Dept(deptno, dname)
Location(locationid, city)
DL(deptno, locationid)
EMP(empno, ename, salary, job, deptno, locationid)
i) Give the name of the employees whose office location is 'Mumbai'.
ii) Give the name of the employess who are in same job as of Tom.
iii) Give the department name which has office in all the locations.

[(CO4)(Apply/IOCQ)]

(c) What is the data type of DUAL table? When this table is required.

[(CO4)(Remember/LOCQ)]

3 + (3 × 2) + 3 = 12

24. A publishing company produces books on various subjects. The books are written by authors who specialize in one particular subject. The company employs editors who, not necessarily being specialists in a particular area, each take sole responsibility for editing one or more book publications. Every book requires some items for publication. These items supplied by suppliers. One supplier can supply many items. Shop owner buys books from the publisher. Shop owner can buy many books but one book can be bought by one shop owner only. Books are uniquely identified by Bookid.

(i) Based on the description above, construct a clean and concise ER diagram for the publishing company database.

[(CO2)(Analyse/IOCQ)]

(ii) Convert the ER design into a set of tables.

[(CO4)(Apply/IOCQ)]

(iii) Merge and reduce the set of tables without introducing redundancy, if possible.

[(CO2)(Apply/IOCQ)]

(6 + 4 + 2) = 12

25. (a) Compare the role of SMQTT and AMQP in the design of IoT.

[(CO1/CO6)(Analyse/IOCQ)]

(b) How do the different functional/topological entities form the device and gateway domain in ETSI M2M?

[(CO6)(Understand/IOCQ)]

(c) Why is UDP called unreliable protocol? *[(CO6)(Understand/LOCQ)]*

4 + 6 + 2 = 12

GROUP – D

26. Write short notes on
    (i) Convolution Code
    (ii) State Diagram
    (iii) Maximum likelihood decoding in convolution code.

[(CO5, CO6)(Remember/LOCQ)]

(3 × 4) = 12

27. (a) For a (6, 3) systematic linear block code, the three parity check bits c_4, c_5 & c_6 are formed following the equations:
$C_4 = d_1 \oplus d_3$, $C_5 = d_1 \oplus d_2 \oplus d_3$, $C_6 = d_1 \oplus d_2$. [Where C & d have their usual meaning]
    i) Write down the generator matrix.
    ii) Construct code words for $i_1 = [101]$, $i_2 = [110]$, $i_3 = [111]$.

[(CO3)(Evaluate/HOCQ)]

(b) Suppose that the received word is 010111. Decode this received word by finding the location of the error if any and the transmitted data.

[(CO4)(Analyse/IOCQ)]

(3+3) + 6 = 12

28. (a) For a (2,1,2) convolution code, $g^0 = (101)$ and $g^1 = (100)$. Draw the encoder. Model the trellis diagram for four successions. *[(CO5,CO6)(Apply/IOCQ)]*

(b) Find the code words for input sequence u = (101) from the trellis diagram constructed above. *[(CO5,CO6)(Analyse/IOCQ)]*

(3 + 7) + 2 = 12

29. (a) For a code vector X=(0 1 1 1 0 0 0) and the parity check matrix H given below, prove that XH^T=(0,0,....,0)

$$H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

[(CO3)(Analyse/IOCQ)]

(b) Given the (7, 3) linear code with generator matrix

$$G = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(i) Reduce G to systematic form
(ii) Find a parity – check matrix for the code *[(CO4)(Analyse/IOCQ)]*

6 + 6 = 12

30. (a) What are the basic differences of EJB and Java Bean?

[(CO6)(Remember/LOCQ)]

(b) Can all application servers run EJB? State when to use EJB.

[(CO6)(Remember/LOCQ)]

(c) What are the main parts of EJB Technology, explain them in brief.

[(CO6)(Remember/LOCQ)]

4 + 4 + 4 = 12

31. (a) Describe the multiplicity in container-managed relationships in the context of Entity Beans.

[(CO6)(Remember/LOCQ)]

(b) Discuss the importance of abstract schema in the context of Entity Beans.

[(CO6)(Remember/LOCQ)]

(c) What makes Message-Driven Beans different from Session and Entity Beans.

[(CO6)(Remember/LOCQ)]

4 + 4 + 4 = 12

32. (a) Explain the operation of a single phase half wave controlled rectifier with RLE load. Draw relevant waveforms. *[(CO2)(Understand/LOCQ)]*

(b) A single phase transformer with secondary voltage 230V, 50 Hz supplies through a half wave rectifier circuit and delivers a power to a load of 15Ω. For a firing angle delay of 45 degrees, calculate (i) Rectification Efficiency (ii) Form Factor (iii) Voltage Ripple Factor (iv) Transformer Utilization Factor (v) PIV of thyristor. *[(CO2)(Evaluate/HOCQ)]*

5 + (3 + 1 + 1 + 1 + 1) = 12

33. Derive the following from the given fuzzy sets A, B, C and D:

$\mu_A(x) = \{(0.5,x_1),(0.8,x_2,),(0.6,x_3),(0.3,x_4)\}$

$\mu_B(x) = \{(1.0,x_1),(0.8,x_2,),(0.0,x_3),(0.1,x_4)\}$

$\mu_C(x) = \{(0.3,x_1),(0.4,x_2,),(0.0,x_3),(0.8,x_4)\}$

$\mu_D(x) = \{(0.3,x_1),(0.7,x_2,),(0.3,x_3),(0.8,x_4)\}$

(i) $\mu_{C\cup B}(x)$ and $\mu_{A\cap C}(x)$

(ii) $\mu_{0.5*D^c}(x)$

(iii) $\mu_A{}^2(x_2)$, $\mu_C{}^3(x_3)$

(iv) $\mu_{D-B}(x)$

(v) $\mu_{B\oplus C}(x)$ *[(CO2)(Analyse/IOCQ)]*

[(1 + 1) + 2 + (1 + 1) + 3 + 3] = 12

34. (a) Suppose a genetic algorithm uses chromosomes of the form $x = a\ b\ c\ d\ e\ f\ g\ h$ with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as: $f(x) = (a + b) - (c + d) + (e + f) - (g + h)$, and let the initial population consist of four individuals with the following chromosomes:

x_1 = 6 5 4 1 3 5 3 2

x_2 = 8 7 1 2 6 6 0 1

x_3 = 2 3 9 2 1 2 8 5
x_4 = 4 1 8 5 2 0 9 4
Evaluate the fitness of each individual, showing all your workings, and arranging them in order with the fittest first and the least fit last.

[(CO6)(Analyse/HOCQ)]

(b) Analyze the following crossover operations:
(i) Cross the fittest two individuals using one–point crossover at the middle point.
(ii) Cross the second and third fittest individuals using a two–point crossover (points b and f).
(iii) Cross the first and third fittest individuals (ranked 1st and 3rd) using a uniform crossover.

[(CO6)(Analyse/IOCQ)]

(c) Suppose the new population consists of the six offspring individuals, received from the crossover operations of the above question. Derive the fitness of the new population, showing all your workings. Has the overall fitness improved? *[(CO6)(Analyse/IOCQ)]*

(d) By looking at the fitness function and considering that genes can only be digits between 0 and 9 choose the chromosome representing the optimal solution (i.e. with the maximum fitness). Determine the value of the maximum fitness. *[(CO6)(Understand/LOCQ)]*

(e) By looking at the initial population of the algorithm, Examine whether it will be able to reach the optimal solution without the mutation operator.

[(CO6)(Understand/LOCQ)]

2 + 3 + 3 + 3 + 1 = 12

35. A genetic algorithm develops population/ chromosomes of the form t = a b c x y z with a fixed length of six genes. Each gene can be any digit between 0 and 7. The proposed fitness function of individual chromosomes is calculated from the relation:
f(t) = (a + b + c) − (x + y) + z
Answer the following questions for the above problem statement
(i) Generate six chromosomes as a population.
(ii) Calculate fitness values
(iii) Apply single-point crossover between 1st and 3rd fittest chromosomes at the midpoint.
(iv) Apply multipoint crossover (at b, x and z position) between 2nd and 4th ranked chromosomes.
(v) What is the maximum fitness value of the chromosome possible in this setup?

[(CO6)(Analyse/IOCQ)]

[2 + 3 + 2 + 3 + 2] = 12

36. (a) What are the driving factors behind the push for refinery-petrochemical integration? *[(CO3,4)(Analyse/IOCQ)]*

(b) A recent trend of Indian refineries is to set up green hydrogen plants in each refinery. Discuss the reasons. *[(CO3,4)(Analyse/IOCQ)]*

6 + 6 = 12

37. Discuss in brief the concept of corporate strategy and its implications in business growth of an organization. *[(CSBS4101.4)(Analyse /LOCQ)]*

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GROUP – E

38. (a) Identify any two scopes to apply sustainable practices in Heritage campus. Analyse how your suggestion is going to help the college (students, staff, teachers, and surrounding community) to get benefitted.

[(CO4)(Analyse/HOCQ)]

(b) Often a smaller, seemingly less important step turns out to be the most significant. Suggest any four ways, how as individuals can we bring smaller changes in our daily life to meet the bigger goal of sustainability? Use examples as required.

[(CO5)(Remember/LOCQ)]

(2x4) + 4 = 12

39. (a) Evaluate $\int_C \{(xy^2 + x)dx + xy\, dy\}$ where C is the closed curve of the region bounded by $y = x^2$and $x = y^2$.

[(MTH1101.6)(Remember/LOCQ)]

(b) Evaluate $\iint_R y\, dxdy$ where R is the region bounded by $y = x$ and the parabola $y = 4x - x^2$.

[(MTH1101.6)(Evaluate/HOCQ)]

6 + 6 = 12

40. (a) Consider the function $f(x, y) = \frac{x^2y^2}{x^2y^2+(x^2-y^2)^2}$. Show that the double limit of $f(x, y)$ does not exist as $(x, y) \to (0, 0)$.

(b) Show that $f(x, y) = \begin{cases} \frac{2xy}{x^2+y^2}, & (x, y) \neq (0,0) \\ 0, & (x, y) = (0,0) \end{cases}$ is not continuous at $(0, 0)$.

[(MTH1101.6)(Analyse/IOCQ)]

(c) Use Divergence theorem to evaluate $\iint_S \{xz^2dy\, dz + (x^2y - z^3)dz\, dx + (2xy + y^3z)dx\, dy\}$ where S is the surface of the hemispherical region bounded by $z = \sqrt{a^2 - x^2 - y^2}$ and $z = 0$.

[(MTH1101.6)(Evaluate/HOCQ)]

3 +3 + 6 = 12

41. (a) If $u = f(y - z, z - x, x - y)$, then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.

[(MTH1101.6)(Understand/LOCQ)]

(b) Show that $\int_C \frac{x^2dy-y^2dx}{x^{\frac{5}{3}}+y^{\frac{5}{3}}} = \frac{3\pi}{16}a^{\frac{4}{3}}$, where C is the quarter of the astroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ in the first quadrant.

[(MTH1101.6)(Analyse/IOCQ)]

6 + 6 = 12

42. (a) If $f(x,y) = \begin{cases} xy\frac{4x^2+5y^2}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$, then prove that $f_{xy} \neq f_{yx}$ using first principle.

[(MTH1101.6)(Analyse/IOCQ)]

(b) Verify Green's theorem for $\oint_C \{(3x-8y^2)dx + (4y-6xy)dy\}$ where C is the boundary of the region bounded by $x = 0$ and $y = 0$ and $x + y = 1$.

[(MTH1101.6)(Apply/IOCQ)]

6 + 6 = 12

43. (a) Evaluate $\int_C (2x+y^2)dx + (3y-4x)dy$ where C denotes the ΔPQR formed by the points $P(0,0)$, $Q(2,0)$ and $R(2,1)$ traversed in the clockwise direction.

[(MTH1101.6)(Evaluate/HOCQ)]

(b) If $z = z(u,v)$, where $u = lx + my$ and $v = ly - mx$, then prove that
$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (l^2+m^2)\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right)$.

[(MTH1101.6)(Understand/LOCQ)]

6 + 6 = 12

44. (a) Evaluate $\int_0^a \int_0^x \int_0^y x^3 y^2 z \, dx \, dy \, dz$.

[(MTH1101.6)(Evaluate/HOCQ)]

(b) Using Stoke's theorem to evaluate $\int_C \{(x+y)dx + (2x-z)dy + (y+z)dz\}$, where C is the boundary of the triangle with vertices $(2,0,0), (0,3,0), (0,0,6)$.

[(MTH1101.6)(Evaluate/HOCQ)]

5 + 7 = 12

45. (a) If $f(x,y) = x^2\tan^{-1}\left(\frac{y}{x}\right) - y^2\tan^{-1}\left(\frac{x}{y}\right)$ then show that $f_{xy} = f_{yx}$.

[(MTH1101.6)(Understand/LOCQ)]

(b) Apply Green's theorem to find $\oint_C \{(\cos x \, siny - xy)\}dx + \sin x \cos y \, dy\}$, where C is the circle
$x^2 + y^2 = 1$.

[(MTH1101.4)(Apply/IOCQ)]

6 + 6 = 12

46. (a) If $u = xf\left(\frac{y}{x}\right) + g\left(\frac{y}{x}\right)$, then show that
i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = xf\left(\frac{y}{x}\right)$

ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0.$

[(MTH1101.6)(Remember/LOCQ)]

(b) Evaluate $\int_0^a \int_0^{\sqrt{x^2-y^2}} (x^2 + y^2) dx\, dy$ by changing to polar coordinate.

[(MTH1101.6)(Evaluate/HOCQ)]

6 + 6 = 12

47. (a) Show that the double limit of $f(x, y)$ does not exist as $(x, y) \to (0, 0)$ where $f(x, y) = \frac{xy^3}{x^2+y^6}$.

[(MTH1101.6)(Analyse/IOCQ)]

(b) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, then prove that

i) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$

ii) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$

[(MTH1101.6)(Remember/LOCQ)]

4 + (4 + 4) = 12